Yoav Grinberg

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Education

University of Twente – BSc in Mechanical Engineering — Sept 2024 – Present
Weighted average grade: 7.4/10
UWC Maastricht – International Baccalaureate Diploma — Sept 2022 – May 2024
Higher Level: Mathematics AA, Physics, Music

Engineering Projects

Self-Balancing Bicycle (Reaction-Wheel Stabilization) — Jan 2026 – Present

- Designing and building a reaction-wheel-stabilized bicycle that balances itself at zero forward speed.
- Iterated the physical mounting system to achieve a rigid platform that is reliable under high-speed operation of the flywheel.
- Integrated a VESC-controlled motor, pulley transmission, and flywheel system able to generate enough reaction torque to return the bicycle back towards upright in tipping testing.
- Debugged stiffness, vibration, alignment, and mounting issues and iterated upon issues.
- Sourced all mechanical and electrical components independently, including motor, VESC, transmission parts, and structural parts.
- Documented decisions, rationales, and failures of the ongoing process on a public technical blog: yoavgrinberg.com/ygbike.

Experience

Founder, ScrollStreet — Nov 2025 – Present

- Built and launched a consumer iOS app, managing product design, development, launch, analytics, and iteration independently.
- Reached 275+ organic users within 72 hours with €0 marketing spend.
- Achieved 39.2% day-1 retention and converted free users to paid customers.

Private STEM Tutor, Freelance — Aug 2025 – Present

- Tutor IB Higher Level physics and mathematics, explaining technical concepts clearly and adapting to learning styles.
- Built a consistent client base through word of mouth referrals.

Skills

Mechanical: SOLIDWORKS, Fusion 360, mechanical prototyping, structural iteration, rotating systems
Mechatronics: ESP32, VESC, soldering, electronics integration, sensors
Programming: Python, MATLAB